← Back to Unit 2 Index

QUESTION 7 • UNIT 2 – NLP / SENTIMENT ANALYSIS

Explain Sentiment Analysis and its types in detail.

📝 5 Marks

Sentiment Analysis & Its Types

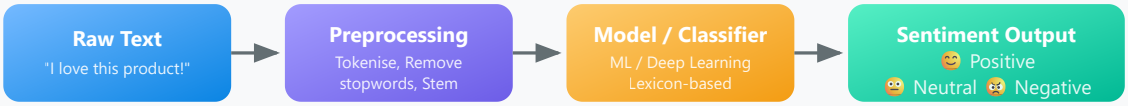
Opinion mining — definitions, types, and diagrams

What is Sentiment Analysis?

Sentiment Analysis (also called *opinion mining*) is a technique in **Natural Language Processing (NLP)** that analyses text to determine the sentiment or emotional tone behind it — whether it is **positive, negative, or neutral**.

It is widely used in customer feedback analysis, social media monitoring, brand reputation tracking, and market research.

Figure 1 — Sentiment Analysis Pipeline



Text goes through preprocessing, a classifier model, and produces a sentiment label as output.

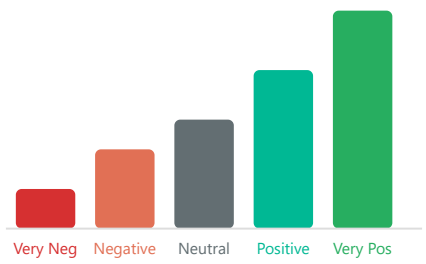
Types of Sentiment Analysis

1. Graded Sentiment Analysis

Goes beyond simple positive/negative. Assigns a **degree of polarity** to the sentiment across 5 levels.

> *"I absolutely love this!" → Very Positive*
> *"The product is okay." → Neutral*
> *"I hate this!" → Very Negative*

Use case: interpreting 5-star ratings or surveys.



5-level polarity scale

2. Emotion Detection

Identifies **specific emotions** such as happiness, anger, sadness, or frustration using lexicons or ML models.

> *"I am so happy!" → Happiness*
> *"This is frustrating!" → Frustration*

Use case: customer service, mental health analysis.

Emotion wheel

3. Aspect-Based Sentiment Analysis

Identifies sentiments related to **specific aspects or features** of a product or service rather than overall sentiment.

> *"Battery life is too short." → Negative (battery)*
> *"Camera quality is amazing!" → Positive (camera)*

Use case: product reviews, feature-level analysis.

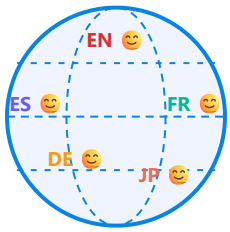
Per-feature sentiment labels

4. Multilingual Sentiment Analysis

Analyses sentiments expressed in **multiple languages**. Requires language detection, translation/multilingual models, and language-specific lexicons.

> *"Este producto es increíble." (Spanish) → Positive*

Use case: global businesses with multilingual feedback.



Sentiment across languages

5. Fine-Grained Sentiment Analysis

Provides detailed, **fine-grained categories** (Very Positive → Very Negative, often 5 classes like star ratings). Similar to graded but more structured.

> *"Extremely satisfied with service!" → Very Positive (5★)*
> *"Service was okay, not great." → Neutral (3★)*

5-star fine-grained scale

6. Intent-Based Sentiment Analysis

Focuses on identifying the **intent behind the text** — whether the user is interested, urgent, neutral, etc., rather than just polarity.

> *"I need this product immediately!" → Urgent*
> *"I might consider buying this later." → Interested*

Use case: marketing campaigns, customer service, lead generation.

Intent classification from text

Applications of Sentiment Analysis

Customer Feedback

Analyse reviews & surveys to understand satisfaction

Brand Monitoring

Track public sentiment about a brand in real-time

Market Research

Predict trends and consumer behaviour

Social Media

Analyse tweets, posts, comments for public opinion

Healthcare

Assess patient feedback and emotional well-being

Political Analysis

Gauge sentiment about candidates or policies